MAROOCHYDORE
Excellence in Education for All
STATE HIGH SCHOOL

Name:_____

Date: _____ Class Code: _____

Science 10 Inheritance and Ratios

Determining genotype and phenotype of progeny (kids)

A. Purebred x purebred

The allele for attached ear lobes (L) is dominant over unattached ear lobes (l).

Use a Punnet square to determine the percentage and ratios of genotypes and phenotypes in each situation:

1. A father with attached ear lobes (LL) and a mother with unattached earlobes (ll)
2. Both father and mother have attached ear lobes (LL)
3. Both mother and father have unattached earlobes (ll)

B. Purebred x hybrid

Use a Punnet square to determine the percentage and ratios of genotypes and phenotypes in each situation:

1. A father with attached ear lobes (Ll) and a mother with unattached earlobes (ll)
2. A father with attached ear lobes (Ll) and a mother with attached earlobes (LL)

C. Hybrid x hybrid

Use a Punnet square to determine the percentage and ratios of genotypes and phenotypes in each situation:

1. Both mother and father have attached ear lobes (Ll)

Determining genotype and phenotype of parents

The allele for having the ability to roll the tongue (T) is dominant over not being able to roll the tongue (t).

For the cases below, use a Punnet square to determine the genotypes of the parents. In some cases, there is more than one possible answer.

1. In a family with 5 children, 4 are able to roll their tongue and 1 cannot. Both father and mother are able to roll their tongues. What are the possible genotypes of the parents?
2. In a family with 5 children, 3 are able to roll their tongue and 2 cannot. The father can roll his tongue but the mother cannot. What are the possible genotypes of the parents?
3. In a family with 5 children, all are able to roll their tongues. The father can roll his tongue but the mother cannot. What are the possible genotypes of the parents?
4. In a family with 5 children, all are able to roll their tongues. Both father and mother are able to roll their tongues. What are the possible genotypes of the parents?